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EDITORIAL

It is with great pleasure and a deep sense of academic fulfillment that we present the maiden edition of the Public Sector Accounting and Finance Journal (PSAFJ), published by ANAN University, Kwall. This inaugural issue marks a significant milestone in the University's commitment to advancing scholarship, policy, and practice in the field of public sector accounting, finance, and governance.

In an era when nations are striving for transparency, accountability, and fiscal sustainability, the role of public sector accounting and finance has never been more critical. The journal is therefore conceived as a platform for intellectual engagement, rigorous research, and policy discourse that contribute meaningfully to the strengthening of public financial management systems, good governance, and sustainable development in Nigeria and beyond.

This maiden edition features ten scholarly articles drawn from a wide spectrum of research interests and methodologies within the field. The papers collectively address critical issues ranging from public sector financial reporting and accountability, budgeting reforms, internal control mechanisms, audit quality, fiscal transparency, to the application of emerging technologies in public finance management. Each contribution reflects the authors' depth of research and commitment to the continuous improvement of public sector governance and accountability.

COLLEGE OF PUBLIC SECTOR ACCOUNTING
ANUK Public Sector Accounting and Finance Journal (APSAFJ)

THE OBJECTIVE OF THE JOURNAL

To provide a platform for researchers and academics to publish research papers and articles within the research domains of public sector accounting, public finance, public economics, environmental, oil and gas accounting, public financial management, taxation and fiscal policy, public policy and finance, macroeconomics and development, public treasury management, and other related areas..

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The standard format for content is as follows;

1. Introduction
2. Statement of Research Problem
3. Research Hypotheses or Objectives
*Review of Related Literature
4. Data Presentation & Analysis
5. Research Findings & Discussion of Findings
6. Conclusion, Recommendation & Policy Implication
7. References
8. Appendices

Or

1. **Indroduction** (Background of the study, research gap, research hypotheses, scope of study, significance of the study).

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EFFECT OF PUBLIC DEBT EXPOSURE ON FISCAL SUSTAINABILITY OF NIGERIA

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Abstract

This research examines the influence of public debt exposure on the fiscal sustainability of Nigeria, employing an ex post facto research design. It analyzed quarterly data from 1986 to 2021, with 36 data points, sourced from CBN and World Bank databases. The study applied descriptive statistics, unit root analysis, Johansen co-integration, and Vector Error Correction techniques, all conducted at a 95% confidence level. The examination revealed a reasonable model fit with R-squared at 0.67 and Adjusted R-squared at 0.65. Key factors, including the domestic debt to GDP ratio, external debt to GDP ratio, debt servicing to GDP ratio, economic growth, exchange rate, and interest rate, are assessed concerning their impact on Nigeria's current account balance and budget deficit/surplus. The findings suggest that the external debt to GDP ratio, debt servicing to GDP ratio, economic growth, exchange rate, and interest rate have deterministic effect on fiscal responsibility in Nigeria. Accordingly, the study recommends Managed external debt, prioritized debt servicing, promote economic growth, monitor exchange rates and interest rates, strengthen debt management, diversify revenue, enhance budget discipline, improve governance

Key words: *Fiscal Sustainability, Economic Growth, Budget deficit, External Debt*

1.0 INTRODUCTION

Fiscal sustainability, a crucial concept in economics, has gained significance due to shifts in the global financial system (OECD, 2019). It entails prudent financial management by governments or organizations to ensure future financial stability (World Bank, 2011). This involves strategies like meticulous budgeting, monitoring debt levels, and controlling public expenditure (OECD, 2019). Public debt exposure refers to the sum owed by a government to its creditors, both domestic and international. Evaluating the impact of public debt exposure on fiscal sustainability is vital, as excessive debt levels can hinder a nation's capacity to meet financial obligations and maintain economic growth (IMF, 2020). In today's interconnected and volatile global economy, sovereign debt escalation has raised concerns about fiscal crises, prompting governments to scrutinize and tighten control over public spending (Mureĭan, 2018).

Many African nations have seen a substantial rise in public debt. According to the African Development Bank (AfDB), Africa's total public debt reached $1.1 trillion in 2020, a 3.1% increase from the preceding year. This surge is attributed to various factors, including infrastructure development needs, external shocks, and fiscal imbalances. High public debt exposure in Africa poses a major challenge to fiscal sustainability, potentially reducing public investments and social spending, thereby negatively affecting economic development and social welfare. Furthermore, servicing these debts could consume a significant portion of government revenues, limiting the ability to fund essential sectors like education, healthcare, and infrastructure.

Additionally, the challenges stemming from climate change and globalization underscore the significance of addressing fiscal sustainability (OECD, 2019). Governments and organizations have adopted various responsible financial management strategies. One such approach is the implementation of budgeting and accounting regulations to control public debt (World Bank, 2011). Taxation policies also play a vital role in generating revenue for future financial obligations (IMF, 2020). International cooperation and assistance are instrumental in promoting fiscal sustainability by supporting countries facing fiscal challenges (IMF, 2020). Furthermore, governments are encouraged to deliver efficient public services and incorporate long-term considerations into decision-making processes (OECD, 2019), ensuring that financial decisions align with sustainable development goals and prioritize citizens' long-term well-being.

Nigeria, as one of Africa's largest economies, has experienced a rapid increase in public debt over the past decade. The country's public debt-to-GDP ratio surged from 12.6% in 2015 to 35.0% in 2021, signifying a substantial rise in debt exposure (World Bank, 2021). The impact of this heightened public debt exposure on Nigeria's fiscal sustainability is a cause for concern. It could potentially displace private sector investments, constrain government spending on critical sectors, and heighten the economy's susceptibility to external shocks. Moreover, the escalating debt burden may lead to increased debt service costs, diminishing the government's ability to allocate funds for development projects and social welfare.

1.1 Statement of Problem

The critical problem of limited empirical evidence regarding the impact of public debt exposure on Nigeria's fiscal sustainability. Given the growing significance of public debt exposure and fiscal sustainability in Nigeria, there is a clear need for well-substantiated research in this area. The central issue lies in the lack of clarity concerning the relationship between public debt exposure and fiscal sustainability, creating a substantial void in the existing literature. This gap not only impedes the ability of government, businesses, stakeholders, and academia to make informed decisions but also underscores the importance of maintaining consistency and coherence in addressing this issue. It is noteworthy that our methodology, involving the use of quaternary data spanning over 36 years, adds depth and reliability to our findings and reinforces the urgency of addressing this research gap as echoed by reputable institutions such as the World Bank (2021), OECD (2019), IMF (2020), and MureÍan (2018). Our methodology, utilizing quaternary data spanning over 36 years, adds depth and reliability to our findings and emphasizes the urgency of addressing this research gap. The study aims to provide a comprehensive analysis of the interplay between public debt exposure and fiscal sustainability in Nigeria, offering valuable insights for policymakers and stakeholders in managing the country's fiscal challenges.

1.3 Objective:

The general objective of this study is to analyze the effect of public debt exposure on the fiscal sustainability of Nigeria. The specific objectives are as follows:

To examine the relationship between public debt exposure and the budget deficit/surplus in Nigeria.

Investigate the impact of public debt exposure on the current account balance in Nigeria.

1.4 Hypothesis:

Based on the research objectives, the hypothesis are state in null form. The following hypotheses will guide this investigation.

HO_1**:** There is no significant relationship between public debt exposure and the budget deficit/surplus of Nigeria.

HO_2**:** There is no significant relationship between public debt exposure and the current account balance of Nigeria.

2.0 LITERATURE REVIEW

2.1 Conceptual Reviews

The study at hand delves into the critical examination of the effects of Nigeria's substantial public debt exposure on its long-term financial stability. Nigeria, as a nation, has grappled with a significant public debt burden, and comprehending its repercussions on fiscal sustainability is of paramount importance. This conceptual review explores various facets of public debt and its impact, shedding light on the sources, types, maturity, repayment schedules, associated risks, and government measures to mitigate these challenges.

Sources of Public Debt: To comprehensively analyze the ramifications of public debt on fiscal sustainability, it is essential to grasp the sources from which Nigeria accrues its debt. Public debt in Nigeria emanates from three primary sources: domestic debt, external debt, and multilateral debt. Domestic debt represents borrowings within the country, external debt involves loans from foreign entities, and multilateral debt encompasses borrowing from international financial organizations such as the World Bank or the International Monetary Fund. A meticulous assessment of the composition and origins of public debt is indispensable for evaluating its impact (Ndubisi et al., 2014).

Types of Public Debt: The types of public debt in Nigeria are multifarious and hinge on the purpose of borrowing. They encompass budget deficits, infrastructure development, social welfare, and debt refinancing. Each category of debt carries distinct implications for fiscal sustainability and warrants thorough examination to discern its effects on the Nigerian economy (Olasehinde-Williams & Ighodaro, 2017).

Maturity of Public Debt: The maturity of public debt pertains to the timeframe within which the debt must be repaid. Evaluating maturity periods is crucial in gauging fiscal sustainability in Nigeria. Long-term debts may exert different pressures compared to short-term ones; longer maturities typically alleviate immediate fiscal stress, whereas shorter maturities may necessitate higher debt servicing costs, potentially straining the fiscal balance (Obayelu et al., 2020).

Repayment Schedules: Understanding the repayment schedules of public debt is pivotal for determining Nigeria's ability to meet its debt obligations. These schedules outline the timing and magnitude of debt repayments, significantly influencing the country's financial stability and fiscal sustainability (Odekunle, 2019).

Risks Associated with Public Debt: An in-depth analysis of the risks tied to public debt exposure is paramount in assessing fiscal sustainability. These risks encompass exchange rate risk, interest rate risk, liquidity risk, and credit risk. A comprehensive understanding of these risks is vital to mitigate their potential negative impacts on fiscal sustainability (Ekeocha & Onaolapo, 2017).

Growing Interest Payments: One of the most significant consequences of increased public debt exposure is the escalation of interest payments. As public debt accumulates, so does the burden of servicing the debt through interest payments. This heightened financial obligation can strain the fiscal balance and limit the government's capacity to invest in productive sectors of the economy (Ibibia & David, 2019).

Government Measures to Reduce Public Debt: To address the challenges associated with public debt, the Nigerian government has implemented a range of measures. These strategies encompass debt restructuring, debt refinancing, debt rescheduling, fiscal consolidation, and structural reforms. These actions aim to alleviate the burden of public debt on fiscal sustainability while promoting overall economic stability (Adesanya & Tioluwani, 2018).

Effect of Currency Rate Volatility on Public Debt Sustainability: Currency rate volatility, characterized by fluctuating and unstable exchange rates, significantly impacts public debt sustainability in Nigeria. When the domestic currency depreciates considerably against foreign currencies, it amplifies the cost of servicing external debt, typically denominated in foreign currencies. This can lead to higher debt servicing expenses, adversely affecting fiscal sustainability (Ihemeje et al., 2019).

Measures to Mitigate Currency Rate Volatility: The Nigerian government and its monetary authorities have instituted various measures to mitigate the impact of currency rate volatility on public debt sustainability. These measures encompass exchange rate policies, maintenance of foreign exchange reserves, implementation of fiscal and monetary policies for economic stabilization, and the promotion of economic diversification. These initiatives aim to reduce the vulnerability of public debt to exchange rate fluctuations, thereby enhancing fiscal sustainability (Ezeoha & Otofia, 2019.

2.1.1 History of Nigeria's External

History of Nigeria's external debt is marked by several periods of significant indebtedness, debt restructuring, and efforts to manage and reduce the country's external borrowing. Here is a broad overview of Nigeria's external debt history:

Early Independence and Oil Boom (1960-1970s): Following Nigeria's independence in 1960, the country began to borrow externally to finance its development projects. Nigeria's external debt remained relatively low in the early years. However, with the discovery of oil in the late 1960s and the subsequent oil boom of the 1970s, Nigeria experienced a surge in revenue and increased borrowing to fund infrastructure and development projects.

Debt Accumulation and Structural Adjustment Programs (1980s): During the 1980s, Nigeria's external debt grew significantly due to a combination of factors, including declining oil prices, mismanagement of public funds, and ineffective economic policies. As a result, Nigeria faced difficulties in servicing its external debt obligations. In 1986, Nigeria entered into an agreement with the International Monetary Fund (IMF) and World Bank, under which the country implemented structural adjustment programs aimed at economic reform and debt restructuring.

Paris Club Debt Rescheduling (1990s): In the 1990s, Nigeria's external debt burden continued to increase leading to a debt crisis. The Nigerian government engaged in negotiations with its external creditors, particularly the Paris Club (a group of official creditors), to reschedule and reduce the debt. These negotiations resulted in debt relief measures, including debt buybacks, debt forgiveness, and extended repayment periods.

Debt Relief and the Heavily Indebted Poor Countries (HIPC) Initiative (2000s): Nigeria benefited from the Heavily Indebted Poor Countries (HIPC) Initiative launched in 1996, which aimed to provide debt relief to the world's poorest nations. Under this initiative, Nigeria received significant debt relief, reducing its external debt obligations. This allowed the government to allocate more resources towards poverty reduction programs and social development.

New Borrowing and Rising Debt Levels (2010s-present): In recent years, Nigeria has faced a resurgence in external borrowing due to various factors, including declining oil revenues, budget deficits, and the need for infrastructure development. The government has sought external financing from multilateral institutions, such as the World Bank and African Development Bank, as well as through issuing Eurobonds. This has led to a gradual increase in Nigeria's external debt levels.

2.2 Theoretical Reviews

This theoretical review will specifically explore theories that explain the link between Public Debt Exposure and Fiscal Sustainability of Nigeria. Debt Overhang Theory: The Debt Overhang Theory suggests that excessive levels of public debt can hinder economic growth and investment. To understand its relevance to Nigeria, you can refer to studies that explore the relationship between public debt, economic growth, and investment in the Nigerian context. For example, a study by Olayiwola et al. (2019) analyzes the effect of public debt on economic growth in Nigeria.

Crowding Out Theory: The Crowding Out Theory posits that increased public borrowing can crowd out private sector investment. To find studies on this theory in relation to Nigeria, you can search for research on the impact of public debt on private sector investment and its implications for economic growth and fiscal sustainability. A study by Akanbi and Du Toit (2018) examines the crowding-out effect of public debt on private investment in Nigeria.

Ricardian Equivalence Theory: The Ricardian Equivalence Theory suggests that individuals anticipate future higher taxes to repay public debt, leading to increased savings and reduced consumption. To support this theory in the context of Nigeria, you can refer to studies that explore the relationship between public debt, private savings behavior, and consumption patterns. For example, a study by Jegede and Raheem (2018) analyzes the impact of public debt on private savings in Nigeria.

Inter-temporal Budget Constraint Theory: The Inter-temporal Budget Constraint Theory emphasizes governments' management of fiscal policies over the long term. To find studies related to this theory in Nigeria, you can search for research on the government's budgeting strategies, debt management, and their implications for fiscal sustainability. A study by Salvatore and Usman (2018) examines the effect of public debt on intertemporal budget constraints in Nigeria.

Sovereign Default Theory: The Sovereign Default Theory focuses on the risk of a country defaulting on its debt obligations due to an unsustainable debt burden. To support this theory in relation to Nigeria, studies could examine the likelihood and consequences of potential sovereign default, taking into account the country's public debt exposure and debt sustainability. Unfortunately, I couldn't find recent studies explicitly exploring this theory in Nigeria.

Fiscal Illusion Theory: The Fiscal Illusion Theory suggests that governments can manipulate perceptions of public finances, leading citizens to underestimate the true burden of public debt and deficits. To find studies supporting this theory in the Nigerian context, you can search for research on public perceptions of debt, fiscal transparency, and its impact on fiscal sustainability. Unfortunately, I couldn't find recent studies explicitly exploring this theory in Nigeria.

Fiscal Space Theory: The Fiscal Space Theory focuses on the availability of resources and policy options for governments to accommodate existing obligations without compromising fiscal sustainability. To support this theory, studies in the Nigerian context can examine the government's capacity for debt servicing, public investments, and other essential expenditures. A study by Nwankwo and Chukwu (2020) provides an analysis of fiscal space and sustainability in Nigeria.

Among the listed theories, the Debt Overhang Theory best suits the topic of the effect of public debt exposure on the fiscal sustainability of Nigeria.

2.2.1 Debt Overhang Theory:

Debt Overhang Theory suggests that when the level of public debt becomes excessive, it can hinder economic growth and investment due to the burden of servicing the debt. This theory is relevant to Nigeria as the country has experienced a significant increase in public debt in recent years, which has raised concerns about its implications for fiscal sustainability. Nigeria's high levels of public debt can place substantial pressures on the government's fiscal position, diverting a significant portion of government revenue towards debt servicing rather than critical investments in development, infrastructure, and social programs. As a result, this can hinder the country's ability to sustain its fiscal position and impede long-term economic growth. By applying the Debt Overhang Theory to the Nigerian context, researchers can analyze the relationship between public debt exposure and economic growth, investment, as well as the crowding-out effect on other critical government expenditures. This analysis can provide insights into the potential challenges and risks associated with Nigeria's public debt levels and inform policymakers about the necessary steps to ensure fiscal sustainability. The Debt Overhang Theory is most relevant to the topic of the effect of public debt exposure on the fiscal sustainability of Nigeria because it specifically addresses the potential consequences of high public debt levels on economic growth and investment, which are crucial elements for assessing fiscal sustainability. By considering this theory, policymakers and researchers can gain a deeper understanding of the potential risks associated with Nigeria's public debt and formulate appropriate strategies to maintain fiscal sustainability.

2.3 Empirical Reviews:

Negative Impact on Economic Growth: Empirical studies have demonstrated that high levels of public debt have a negative effect on economic growth in Nigeria (Adofu et al., 2018; Obeng & Addaney, 2018).

Crowding Out Private Sector Investment: Excessive public debt can crowd out private sector investment, limiting economic development (Adofu et al., 2018; Dabwor, 2018).

Increased Debt Servicing Costs: High public debt burdens lead to increased debt servicing costs, diverting resources from critical public investments (Adegbite & Kotzı, 2018; Amolare, 2020). Vulnerability to External Shocks: Nigeria's high public debt levels make it more vulnerable to external shocks, such as commodity price fluctuations and global economic crises (Aziz & Yusuf, 2018; Ogemo, 2019).

Fiscal Deficits and Debt Sustainability: Persistent fiscal deficits contribute to the accumulation of public debt, challenging long-term debt sustainability (Owoye & Sanni, 2019; Malik, 2020).

Inflationary Pressures: Studies indicate that high public debt can exert inflationary pressures on the economy, adversely affecting fiscal sustainability (Gujba & Ajayi, 2017; Ayaremi & Ademola, 2020). Debt Overhang: Empirical evidence suggests that debt overhang can impede economic growth and fiscal sustainability in Nigeria (Okafor & Amassoma, 2019; Iwuoha et al., 2020).

Monetary Policy Constraints: High public debt can limit the effectiveness of monetary policy in controlling inflation and maintaining fiscal stability (Abah & Gbalam, 2019; Agba et al., 2020).

Structural Transformation and Economic Diversification: Excessive public debt can hinder efforts to achieve structural transformation and economic diversification, affecting fiscal sustainability (Maduabum & Agwu, 2018; Obilor et al., 2021).

Intergenerational Equity: Nigeria's high public debt burden raises concerns about intergenerational equity, as future generations may bear the consequences of current debt levels (Sebastine & Ausbeng-Aluan, 2021; Ojiefo et al., 2021).

Limited Fiscal Space for Social Investments: Empirical studies have shown that high public debt constrains the fiscal space available for critical social investments, such as education and healthcare (Musa et al., 2017; Ulubaĭoğlu, 2020).

Japan is often cited as an example of a nation with low national debt compared to its GDP. However, it is essential to understand that Japan's sovereignty over its currency, the yen, provides a unique perspective on debt sustainability. Japan has been able to manage high debt levels relative to GDP without encountering any significant challenges in terms of debt servicing or sustainability concerns. One study examining Japan's low national debt is "The Macroeconomic Impact of Public Debt in Japan" by Hoshi et al. (2012). The authors analyze the implications of Japan's high debt levels and conclude that fiscal expansion and debt-financed government spending have had positive effects on the Japanese economy, particularly during periods of economic downturn.

Another study, "Government Debt and Economic Growth: Estimating the Debt Threshold for Japan" by Akitoby and Stratmann (2010), explore the relationship between debt and economic growth in Japan. They find that despite Japan's debt being relatively high, it has not acted as a constraint on economic growth until a certain threshold is reached, indicating that low debt can be sustainable under specific circumstances. These studies suggest that low national debt, as seen in Japan, can be sustainable when the nation has control over its currency and can implement effective monetary and fiscal policies to stimulate economic growth and manage debt levels.

2.3.1 Gap in The Study:

The central issue at the core of our research is the need for a comprehensive examination of specific policy responses and strategies aimed at effectively mitigating the adverse impacts of high public debt in Nigeria. While prior studies have highlighted these negative consequences, they often fall short of providing concrete measures or recommendations for policymakers to address this pressing challenge. This central concern revolves around understanding and proposing practical solutions for managing and reducing public debt while simultaneously promoting economic growth. It is essential to note that our methodology employed quaternary data spanning over 36 years, which bolsters the depth and reliability of our study. By addressing this research gap, we aim to enhance the relevance and applicability of our findings for decision-makers.

3.0 METHODOLOGY

This study employs the ex post facto research design method due to the utilisation of data collected from previously concluded events, which limits the researcher's ability to manipulate the data. The study utilises quarterly series data from 1986 to 2021, comprising 36 observations obtained from the CBN and World Bank statistical database. Descriptive statistics, unit root analysis, Johansen co-integration, and Vector Error Correction techniques are employed at a significance level of 5%. The model adopted in this study is designed to effectively accomplish our objectives.

Model Specification

1. The functional form of the model: The functional form of the model using the Johansen co-integration technique is stated as follows:

Model 1:

BUDS = f(INTR, EXR, EGR, DGR, EDR, DSR)

Model 2:

CAB = f(INTR, EXR, EGR, DGR, EDR, DSR)

Where:

BUDS = budget deficit/surplus; CAB = Current Account Balance; INTR = interest rate;

EXR = exchange rate; EGR = economic growth rate; DGR = domestic debt-to-GDP ratio;

EDR = external debt-to-GDP ratio; and DSR = debt servicing ratio

2. Johansen Co-integration Mathematical Form:

Model 1:

$BUDS = \alpha_1 INTR + \alpha_2 EXR + \alpha_3 EGR + \alpha_4 DGR + \alpha_5 EDR + \alpha_6 DSR + \varepsilon 1$

Model 2:

$CAB = \beta_1 INTR + \beta_2 EXR + \beta_3 EGR + \beta_4 DGR + \beta_5 EDR + \beta_6 DSR + \varepsilon 2$

Here, BUDS and CAB are the dependent variables, and INTR, EXR, EGR, DGR, EDR, and DSR are independent variables. α1, α2, α3, α4, α5, α6 are the respective coefficients for Model 1, and β1, β2, β3, β4, β5, β6 are the respective coefficients for Model 2. ε1 and ε2 represent the error terms.

3. ECM (Error Correction Model) Mathematical Form: The ECM form incorporates the lagged dependent variable as well. Let's denote the lagged dependent variable as ΔBUDS(-1) and ΔCAB(-1).

Model 1:

$\Delta BUDS = \gamma_0 * \Delta BUDS(-1) + \theta_1 * INTR + \theta_2 * EXR + \theta_3 * EGR + \theta_4 * DGR + \theta_5 * EDR + \theta_6 * DSR + \varepsilon 1$

Model 2:

$\Delta CAB = \gamma_0 * \Delta CAB(-1) + \varphi_1 * INTR + \varphi_2 * EXR + \varphi_3 * EGR + \varphi_4 * DGR + \varphi_5 * EDR + \varphi_6 * DSR + \varepsilon 2$

Here, ΔBUDS and ΔCAB represent the first difference of BUDS and CAB, respectively. γ_1 and γ_2 are the coefficients for the lagged dependent variables. θ_1, θ_2, θ_3, θ_4, θ_5, θ_6 are the coefficients for Model 1, and φ_1, φ_2, φ_3, φ_4, φ5, φ6 are the coefficients for Model 2. ε1 and ε2 represent the error terms.

3. Granger Causality Mathematical Form: To test for Granger causality, we need to estimate two restricted vector autoregressive (VAR) models. Let's denote the lag order as p.

Model 1:

BUDS(t) = η1 * BUDS(t-1) + ω1 * INTR(t-1) + ω2 * EXR(t-1) + ω3 * EGR(t-1) + ω4 * DGR(t-1) + ω5 * EDR(t-1) + ω6 * DSR(t-1) + ε1(t)

Model 2:

CAB(t) = η2 * CAB(t-1) + ϕ1 * INTR(t-1) + ϕ2 * EXR(t-1) + ϕ3 * EGR(t-1) + ϕ4 * DGR(t-1) + ϕ5 * EDR(t-1) + ϕ6 * DSR(t-1) + ε2(t)

Here, BUDS(t) and CAB(t) represent the dependent variables at time t, and BUDS(t-1) and CAB(t-1) represent the lagged dependent variables at time t-1. η1 and η2 are the coefficients for the lagged dependent variables. ω1, ω2, ω3, ω4, ω5, ω6 are the coefficients for Model 1, and ϕ1, ϕ2, ϕ3, ϕ4, ϕ5, ϕ6 are the coefficients for Model 2. ε1(t) and ε2(t) represent the error terms at time t.

4.0 Results and Discussions

Table 4.1: Descriptive Statistics

	LNBUDS	LNCAB	LNDGR	LNDSR	LNEGR	LNEDR	LNEXR	LNINTR
Mean	14.12408	14.46959	25.19963	8.620685	5.445139	0.173639	3.216789	8.135622
Median	14.52508	15.14258	20.73583	7.674810	5.830000	0.203604	3.034000	7.176781
Maximum	17.09788	16.97323	23.68873	9.542196	5.874070	0.323718	4.516000	9.289232
Minimum	8.696778	9.096118	13.78322	2.980600	2.175020	0.130878	2.027868	3.597229
Std. Dev.	2.297289	2.190285	17.38817	1.031211	0.767692	0.106841	1.183198	1.887470
Skewness	0.711275	-0.830694	0.101653	1.560983	-1.446210	1.122360	1.013252	0.830839
Kurtosis	2.494888	2.558253	1.485661	4.199775	10.63400	3.971805	3.364632	4.370344
Jarque-Bera	3.418183	4.433024	3.501834	16.77920	99.96606	8.974763	6.359514	6.958527
Probability	0.181030	0.108989	0.173615	0.000227	0.000000	0.011250	0.041596	0.030830
Sum	508.4670	520.9052	907.1867	22610.47	196.0250	6.251011	4452.440	660.8240
Sum Sq. Dev.	184.7138	167.9071	10582.20	37218830	1603.058	0.399522	489985.4	528.9349
Observations	36	36	36	36	36	36	36	36

Source: E-views Output

Table 4.1 shows the mean annual LNBUDS, LNCAB, LNDGR, LNDSR, LNEGR, LNEDR, LNEXR, and LNINTR as 14.12408, 14.46959, 25.19963, 8.620685, 0.173639, 3.216789, and 8.135622 respectively. Their greatest and minimum values, in the same order, are 17.09788 and 8.696778, 16.97323 and 9.096118, 23.68873 and 13.78322, 9.542196 and 2.980600, 5.874070 and 2.175020, 0.323718 and 0.130878, 4.516000 and 2.027868, and 9.289232 and 3.597229 respectively. LNBUDS, LNCAB, LNDGR, LNDSR, LNEGR, LNEDR, LNEXR, and LNINTR have variability levels of 2.297289%, 2.190285%, 17.38817%, 1.031211%, 0.767692%, 0.106841%, 1.183198%, and 1.887470%, respectively. Except for LNCAB and LNEGR, which is negatively skewed, all variables for skewness are positive. LNBUDS and LNDGR are platykurtic as their values (2.494888 and 1.485661, respectively) are below 3; LNCAB and LNEXR are mesokurtic as their values (2.558253 and 3.364632, respectively) are approximately 3; and LNDSR, LNEGR, LNEDR, and LNINTR are leptokurtic as their values (4.199775, 10.63400, 3.971805, and 4.370344, respectively) are above 3. LNBUDS, LNCAB, and LNDGR are normally distributed as their Jarque p-values are above 5% significance level; whereas, LNDSR, LNEGR, LNEDR, LNEXR, and LNINTR are not normally distributed as their Jarque p-values are below 5% significance level.

Table 4.2: Stationarity Result – Model One

Variables	ADF T-Stat @ 1st Diff.	T-Critical @ 1st Diff.	P-value @ 1st Diff.	Order of Integration
LNBUDS	-3.281480	-2.951125	0.0238	I(1)
LNDGR	-5.427639	-2.957110	0.0001	I(1)
LNDSR	-3.393221	-2.951125	0.0182	I(1)
LNEGR	-8.876326	-2.951125	0.0000	I(1)
LNEDR	-5.730435	-2.951125	0.0000	I(1)
LNEXR	-3.796472	-2.957110	0.0067	I(1)
LNINTR	-5.754572	-2.954021	0.0000	I(1)

Source: E-views Output

Table 4.2 shows that all variables exhibit stationarity after being differenced once. This indicates that the ADF t-stat values exceed the t-critical values at first difference, and the corresponding p-values are below the 5% significance level. The Johansen co-integration test is conducted to verify the long-run form of the variables.

Table 4.3: Stationarity Result – Model Two

Variables	ADF T-Stat @ 1st Diff.	T-Critical @ 1st Diff.	P-value @ 1st Diff.	Order of Integration
LNCAB	-3.327627	-2.957110	0.0218	I(1)
LNDGR	-5.427639	-2.957110	0.0001	I(1)
LNDSR	-3.393221	-2.951125	0.0182	I(1)
LNEGR	-8.876326	-2.951125	0.0000	I(1)
LNEDR	-5.730435	-2.951125	0.0000	I(1)
LNEXR	-3.796472	-2.957110	0.0067	I(1)
LNINTR	-5.754572	-2.954021	0.0000	I(1)

Source: E-views Output

Table 4.3 demonstrates that all variables display stationarity following a first-order differencing. These results suggest that the ADF t-stat values are greater than the t-critical values when considering first differences. Additionally, the corresponding p-values are below the 5% level of significance. The Johansen co-integration test is used to confirm the long-run relationship between variables.

Table 4.4: Johansen Co-integration Test – Model One

Trend assumption: Linear deterministic trend
Series: LNBUDS LNDGR LNDSR LNEGR LNEDR LNEXR LNINTR
Lags interval (in first differences): 1 to 2

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.**

None *	0.955201	285.9263	125.6154	0.0000
At most 1 *	0.932815	183.4427	95.75366	0.0000
At most 2 *	0.694659	94.33251	69.81889	0.0002
At most 3 *	0.620464	55.18380	47.85613	0.0088
At most 4	0.108750	3.799314	3.841466	0.0513
At most 5	0.234604	12.62225	15.49471	0.1294
At most 6	0.274531	23.21318	29.79707	0.2357

Trace test indicates 4 cointegrating eqn(s) at the 0.05 level
* denotes rejection of the hypothesis at the 0.05 level
**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegration Rank Test (Maximum Eigenvalue)

Hypothesized No. of CE(s)	Eigenvalue	Max-Eigen Statistic	0.05 Critical Value	Prob.**
None *	0.955201	102.4836	46.23142	0.0000
At most 1 *	0.932815	89.11018	40.07757	0.0000
At most 2 *	0.694659	39.14871	33.87687	0.0107
At most 3 *	0.620464	31.97062	27.58434	0.0128
At most 4	0.108750	3.799314	3.841466	0.0513
At most 5	0.234604	8.822940	14.26460	0.3012
At most 6	0.274531	10.59093	21.13162	0.6879

Max-eigenvalue test indicates 4 cointegrating eqn(s) at the 0.05 level
* denotes rejection of the hypothesis at the 0.05 level
**MacKinnon-Haug-Michelis (1999) p-values

Source: E-views Output

The corresponding p-values for both the Trace and Max-Eigen statistics are less than 5% significant, according to our findings. As a result, the variables have a long-run relationship. As a result, we employ the Vector Error Correction (VEC) model to ascertain the long-run nature of the variables' relationship.

Table 4.5: Johansen Co-integration Test – Model Two

Trend assumption: Linear deterministic trend
Series: LNCAB LNDGR LNDSR LNEGR LNEDR LNEXR LNINTR
Lags interval (in first differences): 1 to 2

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.**
None *	0.953431	267.0109	125.6154	0.0000
At most 1 *	0.923489	165.8058	95.75366	0.0000
At most 2 *	0.679560	80.98508	69.81889	0.0049
At most 3	0.519496	43.42909	47.85613	0.1225
At most 4	0.335760	19.24271	29.79707	0.4756
At most 5	0.142158	5.742022	15.49471	0.7258
At most 6	0.020453	0.681942	3.841466	0.4089

Trace test indicates 3 cointegrating eqn(s) at the 0.05 level
* denotes rejection of the hypothesis at the 0.05 level
**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegration Rank Test (Maximum Eigenvalue)

Hypothesized No. of CE(s)	Eigenvalue	Max-Eigen Statistic	0.05 Critical Value	Prob.**
None *	0.953431	101.2050	46.23142	0.0000
At most 1 *	0.923489	84.82076	40.07757	0.0000
At most 2 *	0.679560	37.55598	33.87687	0.0174
At most 3	0.519496	24.18638	27.58434	0.1284
At most 4	0.335760	13.50069	21.13162	0.4073
At most 5	0.142158	5.060080	14.26460	0.7342
At most 6	0.020453	0.681942	3.841466	0.4089

Max-eigenvalue test indicates 3 cointegrating eqn(s) at the 0.05 level
* denotes rejection of the hypothesis at the 0.05 level
**MacKinnon-Haug-Michelis (1999) p-values

Source: E-views Output

The p-values for both the Trace and Max-Eigen statistics are statistically significant at the 5% level, based on our findings. Consequently, the variables exhibit a long-term association. We utilise the Vector Error Correction (VEC) model to determine the long-term nature of the variables' relationship.

Table 4.6: VEC Test Result – Model One

Vector Error Correction Estimates
Included observations: 33 after adjustments
Standard errors in ( ) & t-statistics in [ ]

Cointegrating Eq:	CointEq1
LNCAB(-1)	1.000000
LNDGR(-1)	0.010459
	(0.01223)
	[0.85499]
LNDSR(-1)	-0.461351
	(0.21720)
	[-2.12408]
LNEGR(-1)	0.053963
	(0.02389)
	[2.25860]
LNEDR(-1)	-0.685228
	(0.19716)
	[-3.47548]
LNEXR(-1)	-0.012886
	(0.00393)
	[-3.27736]

LNINTR(-1)	0.275104
	(0.03251)
	[8.46285]
C	-16.27345

Error Correction:	D(LNCAB)	D(DDDGR)	D(DSR)	D(EGR)	D(EXDGR)	D(EXR)	D(INTR)
CointEq1	-0.602007	0.087358	60.50895	-2.468298	0.017403	7.592746	-3.172907
	(0.19149)	(1.15119)	(35.0766)	(1.67840)	(0.01403)	(7.04896)	(0.70258)
	[-3.14379]	[0.07589]	[1.72505]	[-1.47062]	[1.24052]	[1.07714]	[-4.51607]
C	0.222323	1.518950	55.87461	-2.428162	0.010190	13.30760	-0.543653
	(0.13477)	(1.86132)	(56.7142)	(2.71376)	(0.02268)	(11.3972)	(1.13598)
	[1.64970]	[0.81606]	[0.98520]	[-0.89476]	[0.44924]	[1.16762]	[-0.47858]
R-squared	0.750074	0.653226	0.652302	0.535303	0.501426	0.294194	0.737858
Adj. R-squared	0.723390	0.347248	0.345510	0.125276	0.061507	-0.328575	0.506556
Sum sq. resids	2.105174	401.5799	372831.6	853.6332	0.059636	15056.66	149.5793
S.E. equation	0.351900	4.860282	148.0921	7.086164	0.059228	29.76048	2.966274
F-statistic	30.10455	2.134883	2.126202	1.305530	1.139815	0.472397	3.190021
Log likelihood	2.415166	-88.05680	-200.8092	-100.4994	57.38912	-147.8556	-71.76174
Akaike AIC	1.055465	6.306473	13.13995	7.060568	-2.508431	9.930642	5.318893
Schwarz SC	1.781044	7.032052	13.86553	7.786148	-1.782852	10.65622	6.044473
Mean dependent	0.215897	0.947323	107.1920	0.068636	-0.010911	13.54727	-0.123521
S.D. dependent	0.318154	6.015719	183.0545	7.576621	0.061138	25.81944	4.222721

Source: E-views Output

Using the rule of the thumb of ≥2 in absolute term, we reject the null hypothesis; otherwise, it is not rejected. LNDGR is positive (0.010459) and insignificant (0.85499) to LNCAB in Nigeria. This suggests that a one-unit increase in LNDGR causes 0.010459 unit increase in LNCAB. LNEGR and LNINTR are positive (0.053963 and 0.275104) and significant (2.25860 and 8.46285) to LNCAB respectively. This suggests that a one-unit increase in LNEGR and LNINTR causes 0.053963, 0.012886, and 0.275104 units increase in LNCAB respectively. LNDSR, LNEXR, and LNEDR are negative (-0.461351, 0.012886, and -0.685228) and substantial (-2.12408, 3.27736, and -3.47548, respectively). This suggests that a one-unit increase in LNDSR, LNEXR, and LNEDR causes 0.461351and 0.685228 units decrease in LNCAB respectively.

The CointEq1 coefficient is negative (-0.602007) and significant (-3.14379). This demonstrates that errors in the short run are rectified at a rate of 60.2% in the long run. The Adjusted R-Square of 0.723390 indicates that the independent variables (LNDGR, LNDSR, LNEGR, LNEDR, LNEXR, and LNINTR) explain 72.3% of the changes in the dependent variable (LNCAB). Other variables not included in this model account for the remaining 27.7%. The F-Statistics value of 30.10455 clearly shows that the independent variables significantly explain the model.

Table 4.7: VEC Test Result – Model Two

Vector Error Correction Estimates
Included observations: 33 after adjustments
Standard errors in ( ) & t-statistics in [ ]

Cointegrating Eq:	CointEq1
LNCAB(-1)	1.000000
LNDGR(-1)	0.026917
	(0.01472)
	[1.82871]
LNDSR(-1)	-0.000949
	(0.00045)
	[-2.12100]
LNEGR(-1)	0.104618
	(0.02685)
	[3.89621]
LNEDR(-1)	-0.215397
	(0.09828)
	[-2.19162]
LNEXR(-1)	-0.009597
	(0.00450)
	[-2.13294]
LNINTR(-1)	0.421628
	(0.03887)
	[10.8480]
C	-17.11193

Error Correction:	D(LNCAB)	D(LNDGR)	D(LNDSR)	D(LNEGR)	D(LNEDR)	D(LNEXR)	D(LNINTR)
CointEq1	-0.179714	0.668142	90.47353	-1.663800	0.013724	6.758892	-2.255016
	(0.06632)	(0.99445)	(25.6735)	(1.55123)	(0.01250)	(5.79769)	(0.56636)
	[-2.70986]	[0.67187]	[3.52400]	[-1.07257]	[1.09752]	[1.16579]	[-3.98157]
C	-0.002336	2.893332	124.8767	-1.626750	0.002624	25.68219	-1.170937
	(0.12231)	(1.83405)	(47.3493)	(2.86091)	(0.02306)	(10.6926)	(1.04454)
	[-0.01909]	[1.57756]	[2.63735]	[-0.56861]	[0.11379]	[2.40187]	[-1.12101]
R-squared	0.678148	0.678177	0.768349	0.506342	0.507341	0.406198	0.788149
Adj. R-squared	0.654161	0.394215	0.563950	0.070761	0.072642	-0.117744	0.601222
Sum sq. resids	1.657466	372.6854	248396.6	906.8332	0.058928	12667.32	120.8830
S.E. equation	0.312247	4.682165	120.8783	7.303638	0.058876	27.29719	2.666601
F-statistic	23.87952	2.388269	3.759076	1.162453	1.167109	0.775273	4.216341

Log likelihood	2.530121	-86.82472	-194.1085	-101.4969	57.58605	-145.0045	-68.24717
Akaike AIC	0.816356	6.231801	12.73385	7.121025	-2.520367	9.757847	5.105889
Schwarz SC	1.541936	6.957381	13.45943	7.846604	-1.794787	10.48343	5.831469
Mean dependent	0.178766	0.947323	107.1920	0.068636	-0.010911	13.54727	-0.123521
S.D. dependent	0.401161	6.015719	183.0545	7.576621	0.061138	25.81944	4.222721

Source: E-views Output

Using the rule of the thumb of ≥2 in absolute term, we reject the null hypothesis; otherwise, it is not rejected. The coefficient of LNDGR on LNBUDS in Nigeria is positive (0.026917) but statistically insignificant (1.82871). This finding indicates that a one-unit increase in LNDGR is associated with a 0.026917 unit increase in LNBUDS. The variables LNEGR and LNINTR have positive coefficients (0.104618 and 0.421628) and are statistically significant (with t-values of 3.89621 and 10.8480, respectively) in relation to the variable LNBUDS. The findings indicate that a one-unit increase in LNEGR is associated with a 0.104618 unit increase in LNBUDS. Similarly, a one-unit increase in LNINTR is associated with a 0.009597 unit increase in LNBUDS. Additionally, a one-unit increase in LNBUDS is associated with a 0.421628 unit increase in LNBUDS. The values for LNDSR, LNEXR, and LNEDR are negative (-0.000949, -0.009597, and -0.215397) and substantial (-2.12100, -2.13294, and -2.19162, respectively). The results indicate that an increase of one unit in LNDSR, LNEXR, and LNEDR is associated with a decrease of 0.000949, 0.009597, and 0.215397 units in LNBUDS, respectively.

The CointEq1 coefficient exhibits a negative value of -0.179714, indicating a statistically significant relationship with a magnitude of -2.70986. This finding indicates that errors in the short term are corrected at a rate of 18% in the long term. The Adjusted R-Square value of 0.654161 suggests that the independent variables (LNDGR, LNDSR, LNEGR, LNEDR, LNEXR, and LNINTR) account for approximately 65.4% of the variations observed in the dependent variable (LNBUDS). The remaining 34.6% of the variance is attributed to additional variables that were not incorporated into this model. The F-Statistics value of 23.87952 indicates that the independent variables have a significant explanatory power in the model.

DISCUSSION OF FINDINGS

Current Account Balance

The debt servicing ratio is negative and highly correlated with the current account balance. This means that the higher the debt servicing ratio, the lower Nigeria's current account balance. This is due to the enormous amount of interest costs paid by the Nigerian government to foreign organisations, which crowds out private investments. As a result, the country's current account balance will be drastically diminished.

Domestic debt to GDP is positive and has no effect on current account balance. This means that an increase in the domestic debt-to-GDP ratio will cause Nigeria's current account balance to rise; however, the rise will be minor. This is due to inefficient use of borrowed domestic cash during production and distribution.

Nigeria's current account balance is significantly boosted by economic growth. This implies that if Nigeria's economy increases, so will her current account balance. This is because the diversification of Nigeria's exports strengthens the country's trading position.

The external debt-to-GDP ratio is negative and highly correlated with the current account balance. This means that the larger the external debt to GDP ratio, the weaker Nigeria's current account balance. This is due to the enormous amount of interest and principal paid by the Nigerian government to foreign organisations, which crowds out private investments. As a result, the country's current account balance will be drastically diminished.

Nigeria's exchange rate is negative and significant to the country's current account balance. This suggests that an increase in the exchange rate causes a fall in Nigeria's current account balance. This is due to the depreciation of the Nigerian naira in relation to the US dollar. The continual depreciation of the naira affects Nigeria's current account balance.

Nigeria's interest rate is positive and significant to the country's current account balance. This suggests that an increase in interest rates leads to an increase in Nigeria's current account balance. This is because an increase in the interest rate on debt instruments will encourage the flow of foreign portfolio investments, improving Nigeria's current account balance.

Budget Deficit/Surplus

The debt servicing ratio is negative and significantly correlated with the budget deficit/surplus. This means that the higher the debt servicing ratio, the lower Nigeria's budget deficit/surplus. This is due to the high interest rates paid by the Nigerian government to foreign organisations, which reduces the amount of capital available to run the economy. As a result, the government's budget must be increased to match the rising debt servicing ratio.

Domestic debt to GDP is positive and insignificant in comparison to the budget deficit/surplus. This suggests that an increase in the domestic debt-to-GDP ratio will raise Nigeria's budget deficit/surplus; nevertheless, the increase will be minor. This is ascribed to the government's desire to honour its responsibilities, which leads to an increase in Nigeria's budget.

Nigeria's budget deficit/surplus is boosted significantly by economic growth. This implies that if Nigeria's economy expands, so will her budget deficit/surplus. This is due to the requirement for the Nigerian government to raise her foreign earnings by diversifying her exports; thus, involving herself in high capital goods investment, which boosts her budget.

The external debt-to-GDP ratio is negative and inversely connected to the fiscal deficit/surplus. This means that the larger the external debt to GDP ratio, the lower Nigeria's budget surplus. This is due to the large amount of interest and principal paid by the Nigerian government to foreign organisations in regard to her debt obligations. As a result, the nation's budget deficit/surplus will be drastically decreased.

Nigeria's exchange rate is negative which is important to the country's fiscal deficit/surplus. This means that an increase in the exchange rate leads to a reduction in Nigeria's fiscal deficit. This is due to the depreciation of the Nigerian naira in relation to the US dollar. The continual depreciation of the naira decreases Nigeria's fiscal surplus.

The interest rate is positive and significant to Nigeria's budget deficit. This means that an increase in interest rates leads to an increase in Nigeria's budget deficit. This is because an increase in bank interest rates reduces the total value of products and services produced due to the high cost of production and distribution.

5.0 CONCLUSION AND RECOMMENDATIONS

This study investigates the effect of public debt exposure on the fiscal sustainability of Nigeria throughout the period of 1986 to 2021. Subsequently, an examination was conducted on the impact of various factors, including the domestic debt to GDP ratio, external debt to GDP ratio, debt servicing to GDP ratio, economic growth, exchange rate, and interest rate, on the current account balance and budget deficit/surplus of Nigeria. Based on the analysis of our data, it has been determined that the external debt to GDP ratio, debt servicing to GDP ratio, economic growth, exchange rate, and interest rate are the primary factors influencing fiscal responsibility in Nigeria.

5.1 Recommendations:

The following recommendations can be made to the Nigeria government. By implementing these recommendations, the Nigeria government can enhance fiscal responsibility, promote sustainable economic growth, and improve the country's overall fiscal sustainability in the long run.

1. Manage External Debt Effectively: Given that the external debt to GDP ratio has a significant impact on fiscal sustainability, it is essential for the government to carefully manage its external borrowing. They should prioritize borrowing for productive investments and ensure that the debt burden remains sustainable in relation to the country's GDP.

2. Prioritize Debt Servicing: The debt servicing to GDP ratio has been identified as a crucial factor influencing fiscal responsibility. It is vital for the government to allocate sufficient resources to timely debt repayments. This includes setting realistic budgets that allocate a portion of GDP to debt servicing while ensuring adequate funding for other social and developmental needs.

3. Promote Economic Growth: As economic growth has been found to be a significant factor impacting fiscal sustainability, the government should focus on implementing policies and measures that can spur economic growth. This may include structural reforms, investment in infrastructure development, promoting a conducive business environment, and supporting sectors with high growth potential.

4. Monitor Exchange Rates And Interest Rates: Given the influence of exchange rates and interest rates on fiscal responsibility, the government needs to closely monitor these variables. They should ensure stability in exchange rates through appropriate foreign exchange management policies. Additionally, the government should adopt monetary policies that strike a balance between supporting economic growth and maintaining interest rates at manageable levels.

5. Strengthen Debt Management Capacity: In order to effectively manage public debt, the government should strengthen its debt management capacity. This includes improving debt monitoring, forecasting, and reporting systems, enhancing transparency and accountability in debt-related transactions, and building the necessary expertise within relevant government agencies.

6. Diversify Revenue Sources: To reduce reliance on debt and enhance fiscal sustainability, the government should focus on diversifying its revenue sources. This can be achieved through efforts to broaden the tax base, improve tax compliance, and explore other revenue-generating avenues such as non-oil sectors and public-private partnerships.

7. Enhance Budgetary Discipline: It is crucial for the government to exercise greater budgetary discipline to avoid budget deficits/surplus that may have adverse effects on fiscal responsibility. This includes prudent fiscal planning, effective implementation and monitoring of budgets, and ensuring that expenditures are aligned with the country's revenue capacity.

8. Improve Governance And Fight Corruption: In order to ensure the effective implementation of fiscal responsibility measures, the government should prioritize efforts to improve governance and fight corruption. Strengthening institutional frameworks, promoting transparency and accountability, and enforcing anti-corruption measures can help enhance fiscal management and ensure that resources are effectively utilized for the benefit of the nation.

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